ATO CardiTox

Arsenic Trioxide Cardiotoxicity Prediction Report

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I. Background

Arsenic Trioxide (ATO) is a first-line drug for treating Acute Promyelocytic Leukemia (APL). However, ATO may cause cardiotoxicity during treatment, including QT interval prolongation and arrhythmias, which can lead to fatal cardiac events in severe cases. Arsenic undergoes a series of methylation reactions in the body, and the distribution of its metabolites is closely related to the occurrence of cardiotoxicity.

This report uses a deep learning model to comprehensively analyze the patient's arsenic metabolism indicators (iAs, MMA, DMA), metabolic enzyme activity indices (PMI, SMI), and concomitant medication status to predict the risk of ATO cardiotoxicity and provide scientific evidence for clinical decision-making.

II. Test Data

2.1 Input Parameters

Indicator	Value	Unit	Description
iAs	50.00	ng/mL	Inorganic arsenic concentration
MMA	100.00	ng/mL	Monomethylarsonic acid concentration
DMA	100.00	ng/mL	Dimethylarsinic acid concentration
Cardiotoxic Drug	Yes	-	Concurrent use of other cardiotoxic drugs

2.2 Calculated Parameters

Indicator	Value	Percentage	Description
tAs	250.00	100.0%	Total arsenic = iAs + MMA + DMA
iAs%	50.00	20.0%	Inorganic arsenic percentage
MMA%	100.00	40.0%	Monomethylarsonic acid percentage

Indicator	Value	Percentage	Description
DMA%	100.00	40.0%	Dimethylarsinic acid percentage
PMI	2.000	-	Primary methylation index = MMA / iAs
SMI	1.000	-	Secondary methylation index = DMA / MMA

III. Prediction Results

Classification:	**Positive (Toxic)**
Toxicity Probability:	90.8%
Risk Level:	**High Risk**

> **Interpretation:** This patient has a high risk of ATO cardiotoxicity. Close ECG monitoring and treatment adjustment are recommended.

IV. Feature Importance Analysis (SHAP Values)

SHAP (SHapley Additive exPlanations) values explain machine learning model predictions. Positive values indicate the feature increases toxicity risk, while negative values indicate the feature decreases risk. The larger the absolute value, the more significant the feature's impact.

Feature	SHAP Value	Impact Direction	Rank
tAs (Total Arsenic)	0.4089	Increase Risk	1
SMI (Secondary Methylation Index)	0.0540	Increase Risk	2
MMA% (MMA Percentage)	0.0448	Increase Risk	3
DMA% (DMA Percentage)	0.0215	Increase Risk	4
CT_drug (Cardiotoxic Drug)	-0.0657	Decrease Risk	5

> **Major Risk Factor:** tAs (Total Arsenic Concentration)

V. Clinical Recommendations

Based on the prediction results and feature analysis, the following personalized clinical recommendations are provided:

High Total Arsenic Concentration： It is recommended to appropriately reduce ATO dosage or adjust dosing interval, closely monitor ECG QTc interval changes

Concurrent Use of Cardiotoxic Drugs： It is recommended to evaluate the necessity of concomitant medications and adjust the regimen if necessary to avoid additive cardiotoxicity

Low Secondary Methylation Capacity： Patient may have weak arsenic metabolism capacity, it is recommended to strengthen cardiac monitoring frequency

VI. Precautions

Model Characteristics:

This report is generated by a machine learning model and should be used for clinical reference only. It must be combined with clinical examination, laboratory tests, imaging studies, and physician expertise for comprehensive judgment.

Clinical Application:

The prediction model is based on statistical analysis of existing data and cannot cover all individual patient variations. Physicians should make decisions based on the patient's specific condition, medical history, concurrent medications, and other factors.

Monitoring Recommendations:

Regular ECG monitoring and electrolyte testing are recommended throughout treatment. Pay particular attention to QTc interval changes and promptly adjust the treatment plan if abnormalities are detected.

Medication Management:

For patients receiving concurrent medications, carefully evaluate drug interactions. The impact of concurrent cardiotoxic medications on risk varies by individual case and requires comprehensive assessment.

VII. References

1. Zhu J, Chen Z, Lallemand-Breitenbach V, et al. How acute promyelocytic leukaemia revived arsenic. Nat Rev Cancer. 2002;2(9):705-713.

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4. Shen Y, Shen ZX, Yan H, et al. Studies on the clinical efficacy and pharmacokinetics of low-dose arsenic trioxide in the treatment of relapsed acute promyelocytic leukemia. Blood. 1997;89(9):3354-3360.